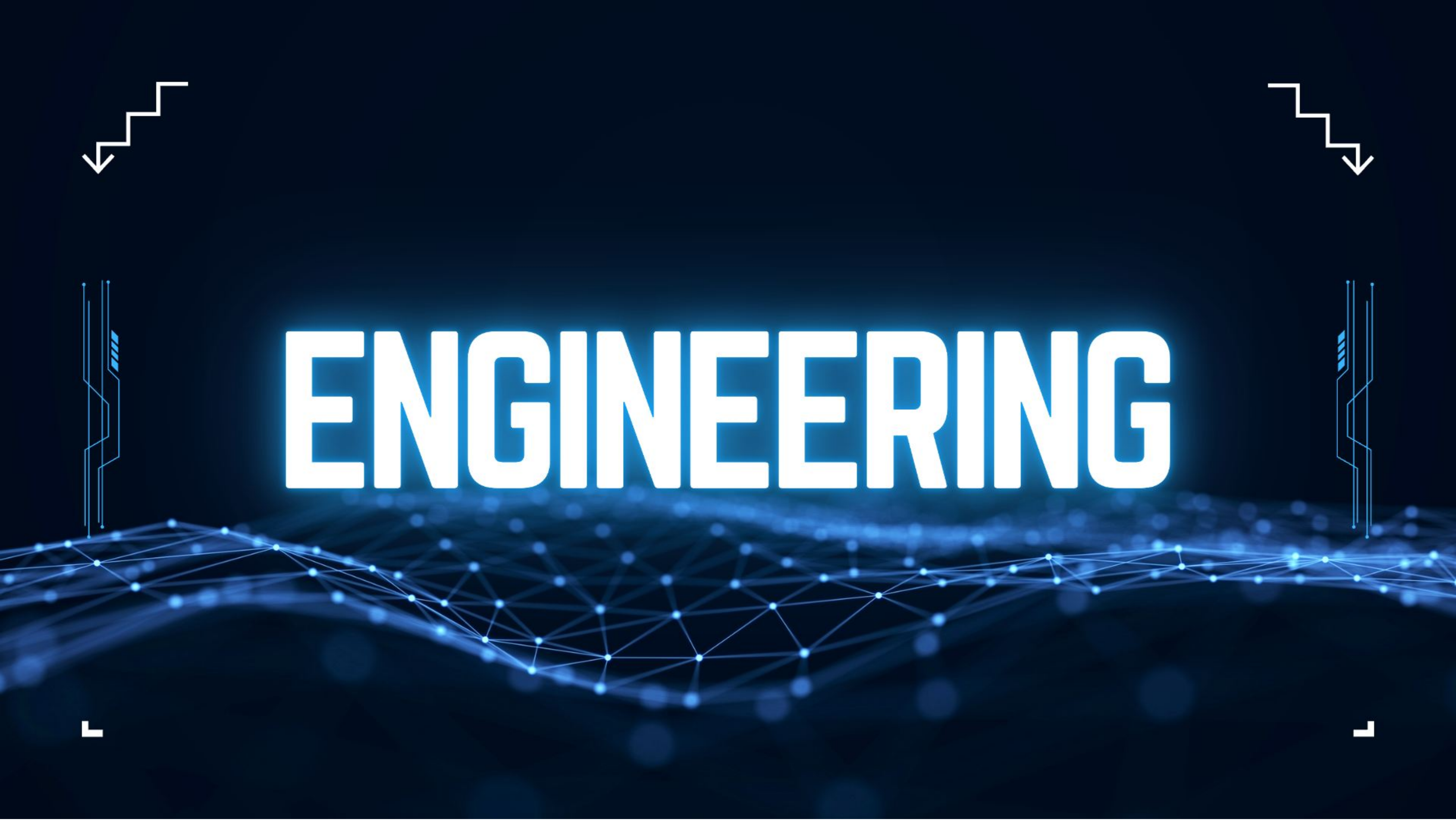
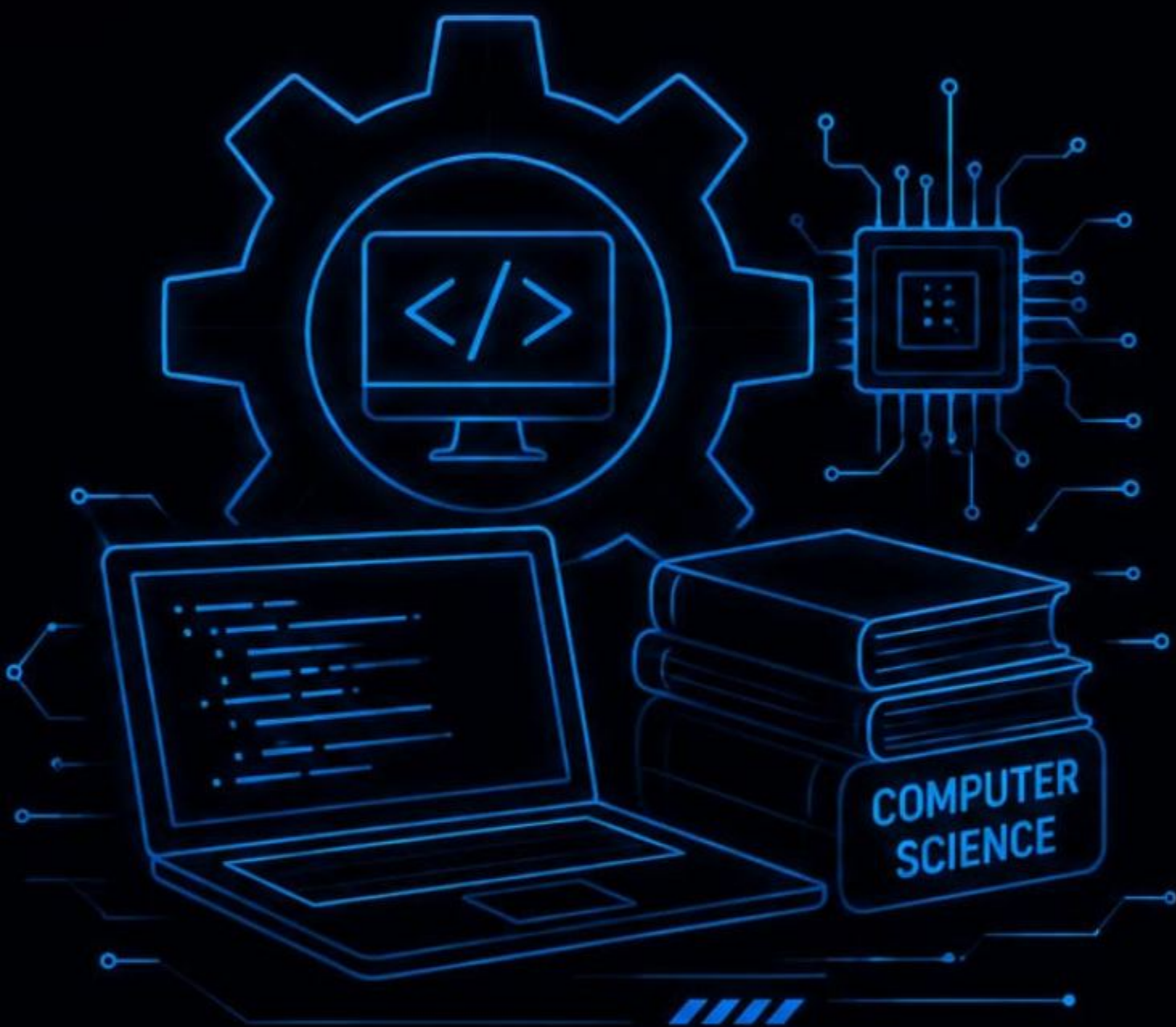




ENGINEERING



ENGINEERING

Engineering is about taking a problem, understanding the constraints (cost, safety, materials, physics), and designing a working solution — then testing and refining it.

SCIENCE VS ENGINEERING

Science

Focuses on understanding why things happen

Discovers laws, facts, and principles of nature

Process: Observation → Hypothesis → Experiment → Theory

Output: Knowledge

Example: Discovering the laws of electricity

Engineering

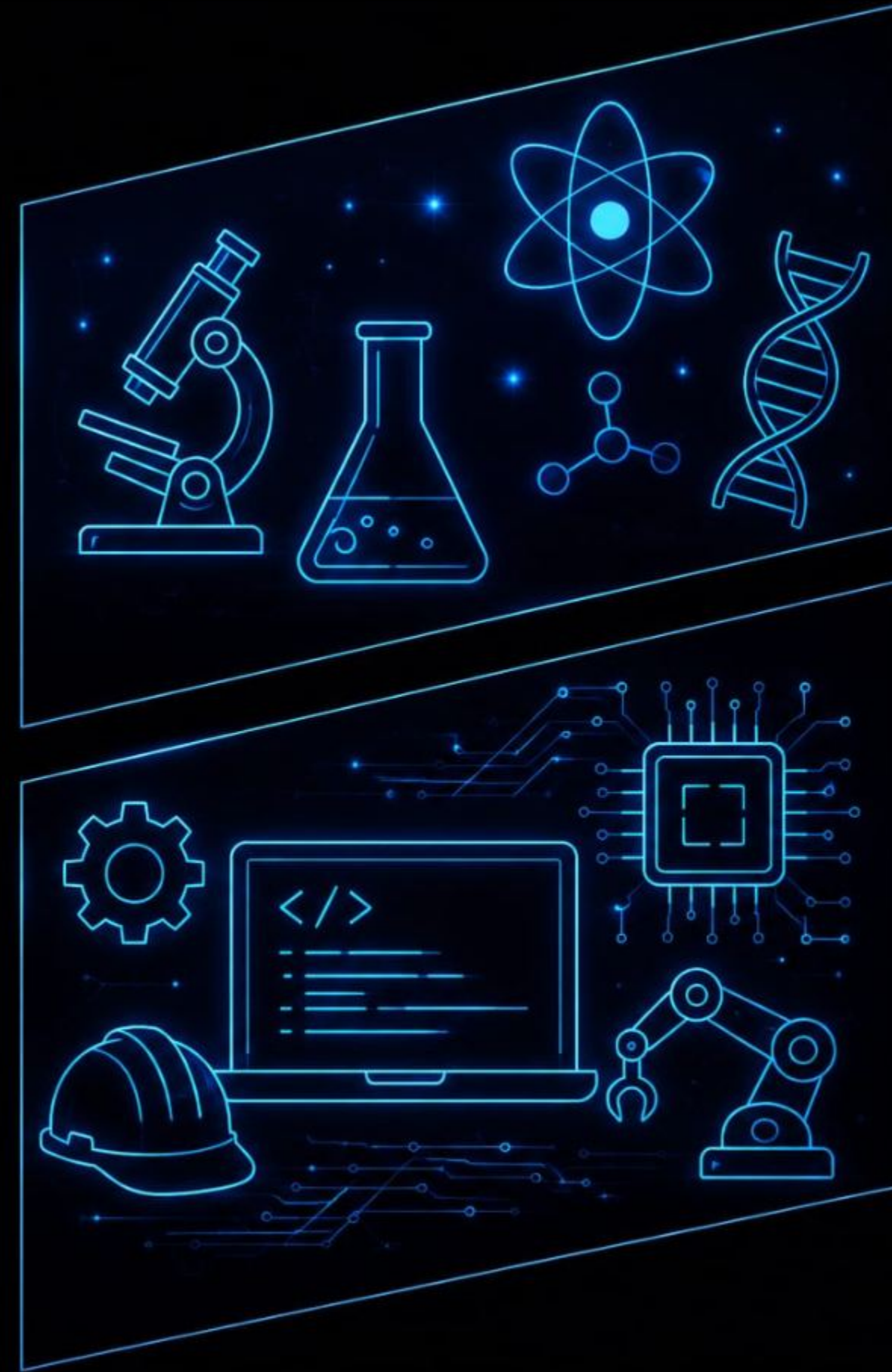
Focuses on how to use that knowledge to solve problems

Applies scientific principles to design and build solutions

Process: Problem → Design → Build → Test → Improve

Output: Products, systems, structures

Example: Using electricity to build power grids and machines



SCIENTIST VS ENGINEER



 The Scientist — "The Explorer"
Curious about the unknown. Asks questions, runs experiments, and uncovers how the world works.

"Why does this happen?"



 The Engineer — "The Builder"
Takes what's already known and turns it into something useful — a machine, a tool, a system.

"How can I make this work?"

NEEDS AND WANTS



NEEDS:

- Strong Programming Skills
- Problem-Solving Ability
- Laptop & Reliable Internet
- Communication & Teamwork

WANTS:

- Industry Certifications
- Internship Experience
- High-Performance Laptop
- AI & Cloud Expertise

MAJOR ENGINEERING DISCIPLINES



 Computer Science & Engineering
(CSE)



 Mechanical Engineering



 Electrical & Electronics Engineering
(EEE)



 Civil Engineering



 Chemical Engineering



 Electronics & Communication Engineering
(ECE)

ROLE OF COMPUTER SCIENCE AND ENGINEERING

CHEMICAL ENGINEERING

Automates industrial processes and analyzes production data
Automated chemical plants, process control systems

ENVIRONMENTAL ENGINEERING

Uses AI and data analytics to monitor pollution and natural resources
Air quality monitoring systems, flood prediction systems



ELECTRICAL ENGINEERING

Develops embedded software and smart power management systems
Smart grids, electric meters

AUTOMOBILE ENGINEERING

Builds software for vehicle control, navigation, and autonomous driving
Electric cars, ADAS, self-driving cars

BIOMEDICAL ENGINEERING

Develops healthcare software and AI-based diagnostic systems
MRI scanners, wearable health trackers



CASE STUDY:UPI-DIGITAL PAYMENTS

SENDING MONEY WITHOUT CASH

Scenario

Before UPI, people had to carry cash or enter bank account details for every transaction. This process was time-consuming and inconvenient.

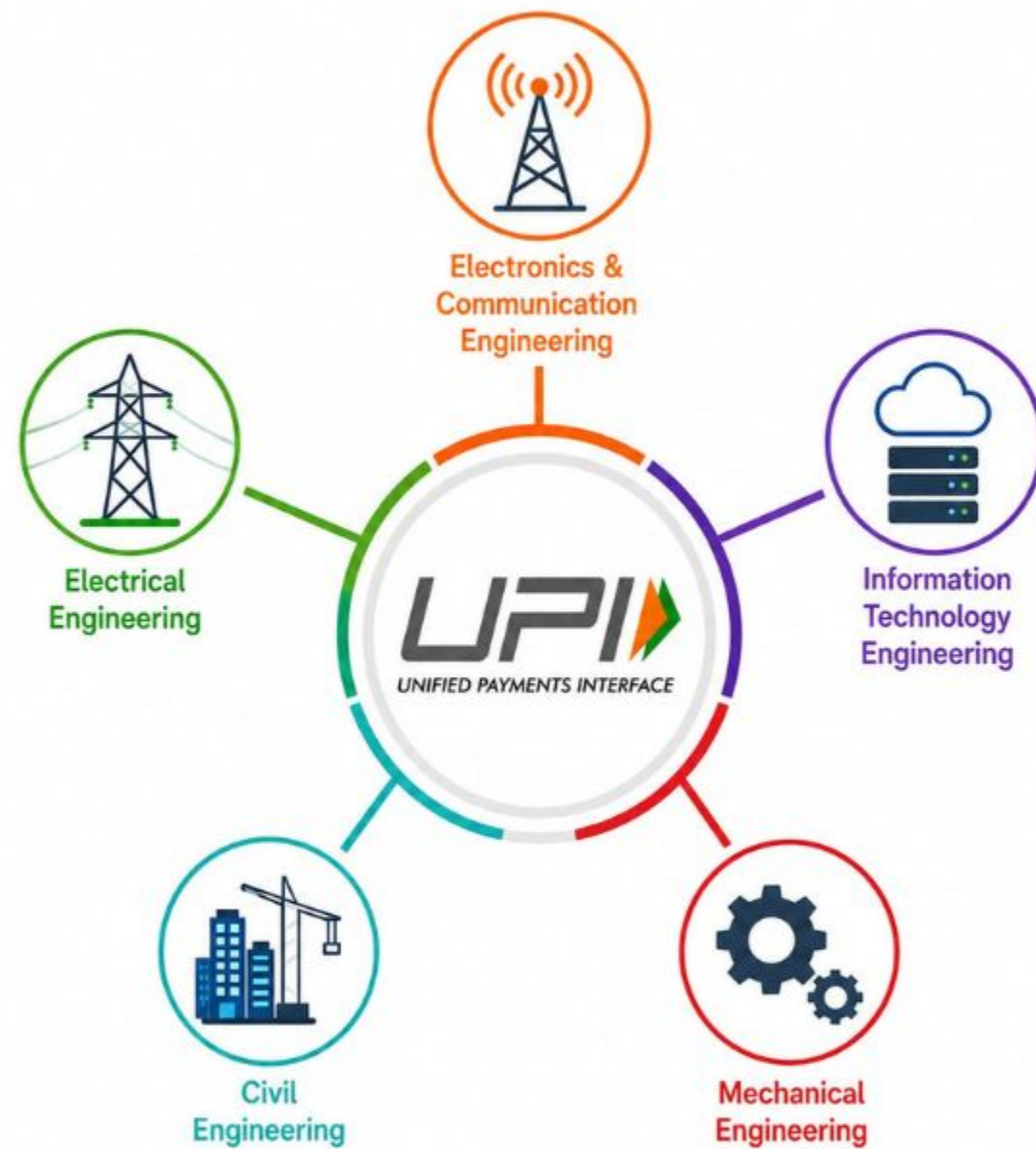


COMPUTER SCIENCE ENGINEERS

- Developed the UPI platform and mobile applications.
- Created secure payment authentication and encryption.
- Built QR code scanning and instant transaction features.
- Managed cloud databases and payment servers.

OTHER ENGINEERING DISCIPLINES

- Electronics and Communication Engineering (ECE) – Designs QR code scanners, payment terminals, and communication hardware used in digital payment systems.
- Electrical and Electronics Engineering (EEE) – Ensures uninterrupted power supply for banks, ATMs, and data centers.
- Information Technology (IT) – Manages network infrastructure, cloud services, and system maintenance for reliable payment processing.
- Civil Engineering – Designs and constructs bank buildings, office spaces, and data centers that support the UPI ecosystem.
- Mechanical Engineering – Designs and maintains ATMs, cash deposit machines, and other banking equipment.





1. "LEARNING ONE LANGUAGE"



2. "COMPUTERS DO ALL THE WORK."



3. "KNOWING PROGRAMMING BASICS MAKES YOU AN EXPERT."

- ❌ Learning one programming language is enough.
- ❌ Computers do all the work.
- ❌ Knowing programming basics makes you an expert.
- ❌ Computer Science is only about coding.
- ❌ CSE professionals only work in IT companies.

MISCONCEPTIONS

SKILLS REQUIRED



PROGRAMMING SKILLS



PROBLEM-SOLVING & CRITICAL THINKING



ARTIFICIAL INTELLIGENCE & MACHINE LEARNING



CYBERSECURITY AWARENESS



CONTINUOUS LEARNING & ADAPTABILITY



AI TOOL USED-CANVA AI



- AI-powered design assistant.
- Creates professional presentations quickly.
- Generates slide layouts and content from prompts.
- Suggests images, icons, and color combinations.
- Saves time and improves presentation quality.
- Easy to customize for users.



Canva

THANK YOU

